

Seamoon Pandey

AI & Backend Engineer

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SUMMARY

AI & Backend Engineer who builds production-grade systems end to end, from fine-tuned transformer models to the distributed services that serve them. Architected an AI-assisted security scanner pairing a DistilBERT classifier with a NestJS microservice backend on a 59k-sample dataset, with strengths across Python ML pipelines, TypeScript/Node backends, and real-time full-stack delivery.

TECHNICAL SKILLS

- **Languages:** Python, TypeScript, JavaScript, Ruby, C/C++, Dart
- **AI / ML:** PyTorch, Transformers (DistilBERT), XGBoost, YOLOv8, ONNX/TorchScript, feature engineering, model evaluation
- **Backend:** NestJS, FastAPI, Express.js, Flask, Ruby on Rails, REST, WebSocket/Socket.IO
- **Data & Infra:** PostgreSQL, Redis, BullMQ, Docker, Docker Compose, microservices, async job queues
- **Frontend:** React, Next.js, Flutter
- **Security:** XSS detection, context-aware payload generation, fuzzing, WAF analysis, headless-browser verification (Playwright)
- **Practices:** system architecture, MVC, auth (JWT/API keys), e2e/unit testing, Git, Linux

FLAGSHIP PROJECT

RedSentinel: AI-Assisted XSS Vulnerability Scanner

Final-year Capstone, Team of 3 | [Code](#) | [Report](#) | [Demo](#)

Lead architect; owned the NestJS orchestration core and the end-to-end ML pipeline. Teammates built the fuzzer and obfuscation engines.

- Architected a polyglot microservice system: a **TypeScript/NestJS** core orchestrating three **Python/FastAPI** AI services over a **Redis/BullMQ** job queue, with **PostgreSQL** persistence and a real-time **Next.js** dashboard (Socket.IO).
- Fine-tuned a **DistilBERT** multi-task classifier to label an injection point's context and severity, replacing brittle signature matching, and reached near-perfect accuracy on an **8,869-sample** held-out split of a deterministic, well-structured label space.
- Built the full ML pipeline on a **59,122-entry** curated XSS dataset: data curation, custom tokenizer training, multi-task training with checkpointing, evaluation, and an inference API with ONNX/TorchScript export and confidence calibration.
- Co-developed an **XGBoost** payload ranker (35 engineered features) to prioritize payloads by predicted effectiveness, with a heuristic fallback when the trained artifact is unavailable.
- Designed the NestJS core: scan lifecycle, crawler, BullMQ queue processor, audit logging, JWT/API-key auth, WebSocket streaming, and HTML/JSON/PDF report generation, backed by unit and e2e test suites.

SELECTED PROJECTS

AI Artifact Recognition Platform

PEC Hackathon | **YOLOv8**, Flask, React

- Built a Flask backend and React dashboard for real-time, YOLOv8-based object recognition with asynchronous inference and structured monitoring endpoints.

Community Connect

OSM Hackathon | Express.js, Flutter

- Developed a full-stack event-coordination platform with REST APIs for real-time GPS-based attendance validation and event tracking.

Garva Diary

BMC InnovateX | Ruby on Rails

- Engineered backend APIs on a Rails MVC architecture with structured validation workflows and mobile integration.

EXPERIENCE

Full-Stack Engineer, Lamida Inc.

Full-time | 23 Feb 2026 – Present

- Build scalable **Web3** applications, including a DAO for the **tUSDT** token.
- **Team Lead**, **Bitfolio** from September 2026.

Freelance Full-Stack Developer

Contract, Remote

- Delivered production full-stack platforms on an **Express.js + React** stack, handling authentication, REST APIs, and third-party integrations:
 - **Jyotisam**: astrology consultation and booking platform (jyotisam.com).
 - **Mahadaan**: donation and charity platform with secure data handling (mahadaan.com).

Mentor, Call for Enthusiasts Program (i-CES)

- Mentored students on building scalable applications with clean backend architecture and development practices.

EDUCATION

Institute of Engineering, Tribhuvan University

2022 to 2026

Bachelor of Engineering in Computer Engineering, Pashchimanchal Campus, Pokhara

Aggregate: **67.98%**, First Division | English: **IELTS Academic 7.5** (CEFR C1), no band below 7.0